NATIONAL ALUMINIUM COMPANY LIMITED (NALCO)

100 Days Quality Management Challenge • Day 84

TECHNICAL REPORT

INTEGRATED ALUMINIUM PROCESS QUALITY ENGINEERING

> Surface Defect Investigation, Process Control, Root Cause Analysis and Continuous Improvement

AUTHOR

Jagriti Singh

DATE

19 October 2026

FOCUS AREA

Primary Aluminium Processing & Surface Engineering Quality Control

Independent Educational Quality Management Report. Compiled for technical evaluation and process improvement study.

1. NALCO & Aluminium Quality Engineering

1.1 About National Aluminium Company Limited (NALCO)

National Aluminium Company Limited (NALCO) is a Navratna Central Public Sector Enterprise (CPSE) under the Ministry of Mines, Government of India. Established in 1981, NALCO is an integrated group with diversified operations in mining, metal, and power. The company operates high-capacity bauxite mines, an alumina refinery at Damanjodi, Odisha, and an aluminium smelter alongside a captive power plant at Angul, Odisha. NALCO produces high-quality primary aluminium products, including ingots, wire rods, billets, cast strips, and rolled products, catering to domestic and international markets (Official website: https://nalcoindia.com/).

1.2 Importance of Quality Engineering in Aluminium Production

Aluminium processing demands strict quality engineering across all production stages—from bauxite extraction and Bayer refining to Hall-Héroult smelting, casting, rolling, and extrusion. Key quality parameters include:

- **Surface Quality:** Prevention of scratches, blisters, oxides, inclusions, mechanical marks, and handling indentations.
- **Dimensional Accuracy:** Tight tolerances on width, thickness, gauge variation, flatness, and profile geometry.
- **Material Properties:** Chemical composition limits (e.g., Fe, Si ratio), grain structure, tensile strength, yield strength, and percentage elongation.
- **Process Consistency:** Thermal management during casting/rolling, bath ratio control in smelting, and controlled cooling.
- **Inspection & Measurement Reliability:** Calibration of optical surface inspection, ultrasonic testing, and dimensional gauges.
- **Traceability & Corrective Action:** End-to-end heat-lot traceability and closed-loop corrective action system (CAPA).

1.3 Quality Beyond Final Inspection

Quality cannot be inspected into a product; it must be built into the manufacturing process. Relying solely on final inspection is inefficient and costly, leading to scrap, rework, delayed deliveries, and uncontained defect risks. Consistent process control, calibrated measurement systems, systematic root cause analysis, and evidence-based decision-making ensure long-term product integrity and operational reliability.

> **Note on Verified Information:** All company organizational details presented above are sourced from public information on NALCO's official website. No access to internal NALCO procedures, private inspection logs, or confidential plant operational data is claimed or implied.

2. Hypothetical Engineering Case Study

2.1 Scenario Description

During routine post-rolling visual and dimensional inspection in a cold rolling mill shop floor, inspection personnel observed recurring longitudinal linear surface marks and fine parallel scratches on cold-rolled aluminium alloy sheets (AA 3003 series, gauge 1.2 mm). The defects appear intermittently across multiple production coils. The exact origin—whether from upstream casting, roller table transfer, roll surface degradation, guide alignment, or post-process handling—is initially unconfirmed.

2.2 Core Investigation Objectives

1. Establish clear problem definition and defect classification parameters.
2. Systematically collect defect frequency, location, and severity data without jumping to early conclusions.
3. Distinguish between physical evidence (e.g., microscopic gouge geometry, transfer marks) and unverified assumptions.
4. Evaluate product conformity against defined client specifications and standard quality threshold limits.

2.3 Potential Contributing Factors for Systematic Examination

- **Material Handling:** Friction or sliding during coil payoff, rewinding, or overhead crane transport.
- **Equipment Contact Surfaces:** Damaged guide rolls, worn stripper plates, accumulators, or debris accumulation on deflectors.
- **Surface Contamination:** Foreign metallic particles, particulate debris in rolling oil, or airborne grit.
- **Operating Parameters:** Strip tension, roll gap alignment, speed transitions, or coolant filtration pressure.

3. Critical Quality Characteristics (CQCs)

3.1 Key Quality Attributes for Rolled Aluminium

Evaluating the impact of a surface defect requires comparing observed characteristics against established design and functional requirements.

Quality Characteristic	Description & Measurement Metric	Engineering Impact
Surface Condition	Absence of cracks, gouges, oxide inclusions, and heavy scratches. Measured by visual rating & surface roughness (R_a).	Affects aesthetic finish, anodizing quality, corrosion resistance, and fatigue strength.
Scratch Depth & Profile	Optical profilometer / depth gauge measurement in micrometers (μm).	Determines whether defect is cosmetic (< 5 μm) or structural notch failure risk (> 25 μm).
Dimensional Tolerances	Thickness uniformity, crown, edge drop, width, and flatness (I-units).	Ensures proper fit-up during customer stamping, forming, or welding operations.
Mechanical Properties	Yield Strength (σ_y), Ultimate Tensile Strength (σ_{uts}), % Elongation.	Governs load-bearing capacity and formability without cracking.

3.2 Defect Classification & Acceptance Criteria

A minor surface mark that does not penetrate protective cladding or exceed allowable depth limits ($t < 2\%$ of sheet thickness) may be acceptable for non-structural painted applications, whereas the same mark would be cause for rejection in architectural anodized or aerospace structural sheets. Acceptance decisions must always align with authorized product specifications rather than subjective opinion.

4. Quality Tools and Practical Application

4.1 Check Sheet for Defect Recording

A check sheet provides a structured, real-time method for collecting qualitative and quantitative defect data at the shop floor level.

Table 4.1: Illustrative Aluminium Surface Defect Check Sheet (Hypothetical Data)

Coil ID	Shift / Date	Defect Type	Location on Strip	Depth (µm)	Observed Frequency	Disposition
C-88401	Shift A / 18-Oct	Linear Scratch	Drive Side Edge (50mm)	12 μm	Continuous line	Segregated / On Hold
C-88402	Shift A / 18-Oct	Handling Impression	Center Width	4 μm	Intermittent (2m pitch)	Accepted (Non-critical)
C-88405	Shift B / 18-Oct	Linear Scratch	Drive Side Edge (52mm)	15 μm	Continuous line	Segregated / On Hold
C-88409	Shift C / 19-Oct	Oxide Inclusion	Random Surface	8 μm	Isolated spots	Rework / Trimmed
C-88412	Shift C / 19-Oct	Linear Scratch	Drive Side Edge (48mm)	14 μm	Continuous line	Segregated / On Hold

4.2 Stratification and Pattern Analysis

Grouping defect data by operational variables highlights strong correlations:

- **By Location:** 82% of linear scratch defects are localized specifically between 45 mm and 55 mm from the Drive Side (DS) edge.
- **By Process Stage:** Defects appear post-entry accumulator table, prior to coiling.
- **By Workstation:** High concentration on Line #2 cold rolling exit section.

Engineering Insight: Spatial concentration near 50 mm DS strongly suggests fixed mechanical contact (e.g., a jammed entry/exit guide strip or burred roller edge) rather than random molten metal inclusions or rolling fluid contamination.

4.3 Statistical Process Control (SPC)

Statistical Process Control distinguishes between common cause variation (inherent system noise) and special cause variation (assignable process disturbances).

> **Key SPC Equations:**
> Upper Control Limit: $UCL = c + 3\sqrt{c}$
> Lower Control Limit: $LCL = max(0, c - 3\sqrt{c})$
> Where c is the average count of nonconformities per sample unit.

If the average defect count $c = 4.0$ per 100m coil length:
$UCL = 4.0 + 3\sqrt{4.0} = 4.0 + 6.0 = 10.0$ defects/coil.
$LCL = max(0, 4.0 - 6.0) = 0$ defects/coil.
Note: Control limits represent process capability, NOT product specification tolerances. An out-of-control signal indicates process instability requiring investigation.

4.4 Root Cause Analysis (5 Whys Application)

Using the 5 Whys method to systematically trace the physical cause of the longitudinal scratch:

1. **Why is there a longitudinal scratch on the coil surface?**
   Physical contact occurred with a rigid stationary object at the exit side guide section.
2. **Why did the coil contact the guide section rigid edge?**
   The drive-side side-trimmer guide plate was misaligned by 4.5 mm relative to strip pass-line.

3. **Why was the guide plate misaligned?**
   The mounting bracket securing bolt had loosened due to line vibration.
4. **Why did the mounting bolt loosen?**
   Standard spring washers were missing after the previous scheduled maintenance overhaul.
5. **Why were spring washers omitted during maintenance?**
   Maintenance reassembly checklist lacked explicit hardware retention inspection steps.

4.5 Failure Mode and Effects Analysis (FMEA)

FMEA prioritizes potential failure modes using Risk Priority Numbers ($RPN = Severity \times Occurrence \times Detection$).

Table 4.2: Illustrative Process FMEA Segment for Aluminium Rolling Exit Stage

Process Step	Potential Failure Mode	Potential Effect	S	Potential Cause	O	Current Controls	D	RPN	Recommended Action
Exit Guide Alignment	Guide plate tilt / misalignment	Linear surface scratch on strip edge	6	Vibration loosening of fasteners	5	Visual inspection end of shift	6	**180**	Install thread-locking hardware & positive mechanical stops.
Coolant Filtration	Filter bypass / tearing	Fine pin-hole denting / debris mark	4	High differential filter pressure	3	Automated DP pressure alarm	3	**36**	Scheduled bi-weekly filter cartridge replacement.
Coiler Payoff	Tension fluctuation / slippage	Interlayer chatter marks / scuffing	5	Load cell sensor calibration drift	3	Routine gauge check	4	**60**	Recalibrate tension load cells monthly.

4.6 Measurement and Inspection Reliability

Variability in defect reporting often stems from inspection methodology rather than actual product changes. Key reliability factors include:

- **Lighting Standards:** Minimum 1000 lux illumination at inspection station with diffusers to prevent glare on reflective aluminium surfaces.
- **Viewing Geometry:** Inspection viewing angle maintained at $45° \pm 5°$ relative to strip travel vector.
- **Optical Measurement:** Depth gauge optical calibration verified against certified master block standards.
- **Inter-Rater Agreement:** Resolving inspector discrepancies through standardized visual boundary samples (Accept/Reject limit boards).

4.7 Corrective Action Framework

A robust resolution requires three distinct action levels:

- **Immediate Correction (Containment):** Quarantining suspected coils (C-88401 to C-88412) and trimming affected edge strips.
- **Corrective Action (Root Cause Removal):** Re-aligning guide plate to exact pass-line, replacing damaged hardware with vibration-resistant locking fasteners.
- **Preventive Action (Systemic Improvement):** Updating preventive maintenance checklists to require torque verification and lock-washer checks.

5. Complete Engineering Investigation Workflow

The structured workflow below ensures evidence-based decision-making during surface quality defect investigations:

1. PROBLEM IDENTIFICATION (Detect defect & freeze immediate process)

↓

2. DEFECT CLASSIFICATION (Characterize geometry, depth, orientation, type)

↓

3. DATA COLLECTION (Log check sheets, batch IDs, shift logs, heat history)

↓

4. PATTERN ANALYSIS (Stratify data by DS/WS edge, time, roll set, operator)

↓

5. INSPECTION VERIFICATION (Verify lighting, optical gauges, inspector consistency)

↓

6. ROOT CAUSE INVESTIGATION (Conduct 5 Whys, fishbone diagram, physical inspection)

↓

7. RISK ASSESSMENT & FMEA (Evaluate product impact, calculate RPN, establish containment)

↓

8. CORRECTIVE ACTION SELECTION (Design mechanical fix, process tuning, hardware upgrade)

↓

9. VALIDATION & IMPLEMENTATION (Execute trial run under controlled observation)

↓

10. EFFECTIVENESS MONITORING (Track defect occurrence across next 30 production batches)

↓

11. DOCUMENTATION & CONTINUOUS IMPROVEMENT (Update SOP, PM logs, FMEA, training modules)

6. Proposed Engineering Solution & Implementation Plan

The proposed solution implements a step-by-step resolution model across twelve distinct stages:

Stage	Action Item	Required Engineering Evidence	Expected Outcome
1. Define Problem	Log exact defect description: longitudinal scratch on AA 3003 sheet.	Defect logging report & photographic record.	Clear, unambiguous problem statement.
2. Classify Defect	Measure scratch depth ($12–15\ \mu m$) and location (50 mm DS edge).	Optical profilometer trace printout.	Defect categorized as mechanical score mark.
3. Collect Data	Deploy standardized check sheet across all rolling lines.	Completed check sheets for 50 consecutive coils.	Full defect frequency dataset.
4. Look for Patterns	Perform stratification analysis by line, shift, and position.	Stratification charts showing 82% DS edge concentration.	Investigation focused on DS exit equipment.
5. Verify Inspection	Calibrate depth gauges and perform gauge R&R study.	Gauge calibration certificates ($\%GRR < 10\%$).	Confirmed measurement system reliability.
6. Investigate Cause	Inspect DS guide assembly; perform 5 Whys inquiry.	Physical finding: loose guide bolt & wear mark on plate.	Confirmed physical root cause.
7. Assess Risk	Evaluate structural/functional impact of $15\ \mu m$ scratch on sheet.	Stress concentration analysis & customer spec review.	Controlled containment & edge slit disposition.
8. Select Action	Redesign mounting hardware with positive locking pins.	Approved engineering design change request (ECR).	Elimination of bolt loosening risk.
9. Implement & Document	Install upgraded guide plates and update PM SOPs.	Work order sign-off & updated SOP document.	Permanent physical & procedural fix.
10. Validate Result	Run a 5-coil test trial under strict monitoring.	Zero scratches observed on trial coils.	Trial validation success.
11. Monitor Effectiveness	Track defect rates on SPC charts for 30 operating days.	P-chart / C-chart showing zero special cause spikes.	Long-term effectiveness verified.
12. Continuous Improvement	Standardize fix across Lines #1, #3, and update FMEA.	Revised FMEA document with RPN reduced from 180 to 24.	Horizontal deployment complete.

7. Risk Assessment

Uncontrolled surface defect recurrence imposes significant technical and commercial risks:

- **Financial Waste & Rework:** High scrap rates from edge slitting or re-melting prime aluminium coils.
- **Customer Dissatisfaction:** Line stoppages at automotive or packaging assembly plants due to feeding jam or surface rejection.
- **Functional Integrity Degradation:** Deep linear scratches ($> 25\ \mu m$) act as stress raisers, initiating micro-cracks during high-strain deep drawing or bending operations.
- **Operational Inefficiency:** Unscheduled line interruptions and prolonged troubleshooting downtime.

8. Validation and Expected Practical Impact

Table 8.1: Corrective Action Validation Matrix

Proposed Action	Evidence Required	Verification Method	Expected Benefit
Guide plate positive lock modification	Mechanical drawing, installation sign-off, torque log	Physical audit & vibration testing	Prevents guide plate drift during high-speed rolling.

Proposed Action	Evidence Required	Verification Method	Expected Benefit
Standardized inspection illumination	Lux meter calibration log (>1000 lux verified)	Light intensity grid audit	Eliminates missed defect detection on reflective strip.
PM checklist update for fasteners	Revised PM instruction sheet #PM-AL-CR-04	Maintenance log audit	Ensures mandatory lock washer inspection during overhauls.
Operator defect classification training	Training attendance log & competency test	Inter-rater agreement check	Achieves 100% consistency in defect logging and disposition.

9. Key Engineering Learnings

1. **Systematic Logging:** Standardized check sheets prevent vague defect descriptions and establish objective baselines.
2. **Data Stratification:** Grouping defect occurrence by physical position immediately narrows down root cause search areas.
3. **Evidence Over Assumptions:** Never implement costly process changes based on speculation without physical or statistical proof.
4. **Measurement System Integrity:** Uncalibrated gauges or inadequate lighting can simulate non-existent process instability.
5. **Root Cause Depth:** Superficial repairs fail if systemic maintenance checklist gaps remain unaddressed.
6. **FMEA Prioritization:** Risk Priority Numbers provide a rational basis for capital and engineering resource allocation.
7. **Distinguishing Control vs. Spec Limits:** Control limits reflect process capability; specification limits define product acceptance.
8. **Three-Tier Correction:** Effective resolution requires immediate containment, root cause removal, and systemic prevention.
9. **Follow-up Verification:** An action is not complete until long-term data proves permanent defect reduction.
10. **Process-Embedded Quality:** Long-term quality excellence relies on robust engineering design and process control, not downstream inspection.

10. Technical Interview Questions and Answers

Independently developed educational practice questions for Quality Engineering & Aluminium Manufacturing concepts.

Category 1: Basic Level (Questions 1 to 10)

BASIC **Q1: What is National Aluminium Company Limited (NALCO) and what are its primary manufacturing operations?**

A: NALCO is a Navratna CPSE under the Ministry of Mines, Government of India. It is an integrated aluminium complex operating bauxite mines, an alumina refinery at Damanjodi, and an aluminium smelter with a captive power plant at Angul, Odisha, producing ingots, wire rods, billets, and rolled products.

BASIC **Q2: Why is surface quality particularly critical in rolled aluminium sheet products?**

A: Rolled aluminium sheets are widely used in architectural, packaging, and transport applications where surface defects impair aesthetic anodizing/painting, act as stress concentrators during forming operations, and reduce fatigue resistance.

BASIC **Q3: Explain why product quality cannot be guaranteed by final inspection alone.**

A: Final inspection only detects defects after value has been added; it does not prevent defect occurrence. Relying solely on final inspection leads to high scrap costs, rework, and the risk of uncontained defect escapes if inspection efficiency is imperfect.

BASIC **Q4: What is the primary purpose of a Quality Check Sheet in a manufacturing setup?**

A: A check sheet provides a structured, standardized real-time format for shop floor operators to record defect occurrences, types, locations, and frequencies for subsequent statistical analysis.

BASIC **Q5: Define the 5 Whys technique in Root Cause Analysis.**

A: The 5 Whys is an iterative interrogative technique used to explore the cause-and-effect relationships underlying a problem by repeatedly asking "Why?" until the fundamental systemic root cause is identified.

BASIC **Q6: What is the difference between a cosmetic surface mark and a functional defect on an aluminium coil?**

A: A cosmetic mark is a shallow surface impression ($< 5\ \mu m$) that does not breach protective cladding or impair strength, whereas a functional defect exceeds depth thresholds, compromising structural formability, corrosion resistance, or assembly.

BASIC **Q7: What is the difference between immediate containment and corrective action?**

A: Immediate containment prevents suspected nonconforming product from reaching customers (e.g., quarantining, edge slitting), while corrective action eliminates the underlying root cause to prevent defect recurrence.

BASIC **Q8: What role does lighting intensity play in surface defect inspection of reflective metals?**

A: Inadequate or glare-heavy lighting causes visual fatigue and missed defects. Controlled high-intensity diffused lighting (>1000 lux) at appropriate viewing angles ($45°$) ensures reliable visibility of fine scratch profiles.

BASIC **Q9: Why must physical measurement instruments like depth profilometers undergo periodic calibration?**

A: Calibration establishes instrument accuracy against traceable master standards, ensuring measurement values (e.g., scratch depth in micrometers) are true and free from sensor drift or bias.

BASIC **Q10: What is the function of an inspection disposition log?**

A: An inspection disposition log records official acceptance, scrap, rework, or concession decisions for inspected material, ensuring complete quality traceability and material accountability.

Category 2: Moderate Level (Questions 11 to 20)

MODERATE **Q11: How does defect stratification aid in narrowing down potential root causes during an investigation?**

A: Stratification sorts raw defect data by specific factors (e.g., position across strip width, shift, roll stand). Concentrated defect clusters (e.g., 82% at 50 mm drive side) isolate the fault to specific machinery components rather than random systemic causes.

MODERATE **Q12: Differentiate between Common Cause and Special Cause variation in Statistical Process Control.**

A: Common cause variation is inherent noise stable within a process (e.g., minor temperature flux), requiring system redesign to reduce. Special cause variation stems from specific external disturbances (e.g., a loose guide bolt), requiring targeted corrective intervention.

MODERATE **Q13: Why are Control Chart limits (UCL/LCL) distinct from Product Specification limits?**

A: Control limits are calculated statistically from actual process performance data ($\pm 3\sigma$) to indicate process stability. Specification limits are set by engineering design or customer requirements to define product acceptability.

MODERATE **Q14: Explain the calculation and significance of Risk Priority Number (RPN) in FMEA.**

A: $RPN = Severity\ (S) \times Occurrence\ (O) \times Detection\ (D)$. RPN ranks potential failure modes quantitatively, allowing quality teams to prioritize engineering resources toward high-risk items.

MODERATE **Q15: What steps should be taken if two inspectors give conflicting acceptance decisions on the same aluminium coil?**

A: Re-evaluate the coil using calibrated quantitative instruments (e.g., optical depth profilometer) against authorized limit samples, review standard visual acceptance criteria, and conduct gauge repeatability & reproducibility (GRR) training to align inspectors.

MODERATE **Q16: How can rolling oil filtration failure cause surface defect patterns on aluminium sheets?**

A: Suspended metallic debris or fine particulates in unfiltered rolling fluid become trapped between work rolls and the strip, creating random microscopic denting, pin-hole impressions, or fine abrasive scratches.

MODERATE **Q17: Describe the engineering rationale behind using positive mechanical locking fasteners on high-vibration rolling machinery.**

A: High-speed rolling mills induce continuous cyclical vibration that overcomes friction-based fastener clamping, causing standard bolts to back out. Positive locking mechanisms (e.g., lock wire, safety tabs, thread-locking compounds) physically prevent rotational loosening.

MODERATE **Q18: What is a P-chart, and when is it appropriately applied in quality control?**

A: A P-chart is an attribute control chart used to monitor the proportion (p) of nonconforming items in samples of equal or varying size, suited for tracking overall pass/fail coil yield rates.

MODERATE **Q19: How do you establish whether a surface scratch impairs the forming capability of an AA 3003-H14 sheet?**

A: Perform tensile and bend testing across samples containing the scratch. Measure localized stress concentration, micro-crack initiation under bending radius, and compare remaining yield/ultimate tensile strength against specification standards.

MODERATE **Q20: Why is long-term follow-up monitoring mandatory after executing a corrective action?**

A: Immediate post-fix trial success may be coincidental or influenced by temporary trial conditions (Hawthorne effect). Monitoring production over 30+ operating days confirms permanent elimination of special cause recurrence under routine operational variability.

Category 3: Advanced Level (Questions 21 to 30)

ADVANCED **Q21: How do you mathematically and physically distinguish correlation from causation during a mill line defect investigation?**

A: Correlation demonstrates a statistical association (e.g., defects increasing during Shift B via high Pearson coefficient r), whereas causation requires demonstrating a physical mechanism (e.g., mechanical contact mark matching guide geometry) validated by controlled hypothesis testing (DOEs) and replication.

ADVANCED **Q22: In a rolling process producing continuous coil strip, why might a standard C-chart be mathematically invalid for modeling scratch defects?**

A: A C-chart assumes a Poisson distribution where defects occur independently across a fixed area. A continuous scratch caused by a fixed mechanical guide is a single continuous event rather than independent Poisson point occurrences, violating independence assumptions.

ADVANCED **Q23: How does Gauge Repeatability and Reproducibility (%GRR) impact Statistical Process Control data integrity?**

A: High %GRR (>30%) indicates that measurement system variation dominates overall observed process variance ($\sigma^2_{total} = \sigma^2_{process} + \sigma^2_{measurement}$), making control chart limits artificially wide and masking actual process shifts.

ADVANCED **Q24: Explain the engineering trade-offs when establishing acceptable scratch depth thresholds for automotive vs. architectural aluminium sheets.**

A: Automotive outer body panels prioritize paint appearance and deep-drawing formability, requiring ultra-stringent scratch depth limits (<2 μm). Architectural structural panels prioritize corrosion resistance and load capacity, permitting deeper cosmetic marks (<25 μm) provided anodizing integrity is maintained.

ADVANCED **Q25: What statistical technique would you employ to evaluate whether a modified roll guide significantly reduced scratch occurrence compared to baseline production?**

A: Use a two-sample hypothesis test (e.g., two-sample proportion test or Mann-Whitney U test depending on data normality) comparing pre-fix and post-fix defect rates to verify statistical significance ($p < 0.05$).

ADVANCED **Q26: If a recurring scratch defect reappears 45 days after an apparently successful corrective action, how should the quality engineering team proceed?**

A: Re-open the investigation without assuming the original root cause diagnosis was incorrect. Examine whether the implemented control degraded (e.g., fastener torque loss, physical wear of modified parts) or if a new secondary failure mode producing identical defect morphology has emerged.

ADVANCED **Q27: How can Failure Mode and Effects Analysis (FMEA) be integrated into a plant's preventive maintenance (PM) management system?**

A: FMEA high-RPN failure modes directly populate PM work orders by defining mandatory inspection tasks, hardware replacement intervals, and critical check metrics (e.g., guide torque verification) based on failure severity and occurrence risk.

ADVANCED **Q28: Discuss the impact of work roll surface roughness (R_a) degradation on aluminum strip surface defect formation during cold rolling.**

A: As work roll surfaces suffer thermal fatigue or localized abrasive wear, their micro-topography transfers directly onto the soft aluminium strip, causing localized friction spikes, micro-galling, surface pick-up, and continuous linear scuffing.

ADVANCED **Q29: What engineering criteria justify accepting a non-conforming aluminium coil under concession (use-as-is)?**

A: Concession requires documented engineering evaluation proving the defect does not compromise customer structural, functional, safety, or aesthetic requirements, backed by formal customer sign-off and authorized quality authority approval.

ADVANCED **Q30: How does robust root cause elimination contribute to overall corporate sustainability and ESG targets in primary aluminium production?**

A: Primary aluminium production is energy-intensive (14-15 kWh/kg). Eliminating internal scrap and re-melting loops drastically reduces electrical power consumption, greenhouse gas emissions, processing chemical consumption, and operational carbon footprint.

11. References

1. **National Aluminium Company Limited (NALCO):** Official Corporate Portal & Overview. Available at: https://nalcoindia.com/
2. **ISO 9001:2015:** *Quality Management Systems — Requirements*. International Organization for Standardization, Geneva, Switzerland.
3. **ISO 31000:2018:** *Risk Management — Guidelines*. International Organization for Standardization, Geneva, Switzerland.
4. **Montgomery, D. C. (2019):** *Introduction to Statistical Quality Control*, 8th Edition, John Wiley & Sons, New York.
5. **ASM Handbook, Volume 2B:** *Properties and Selection: Aluminium Alloys*. ASM International, Materials Park, Ohio.

12. Disclaimer

This is an independent educational case study prepared by **Jagriti Singh** as part of the **100 Days Quality Management Challenge**. The engineering scenario and any illustrative data are hypothetical and do not describe an actual NALCO incident, actual production results, or an internal NALCO procedure. The proposed actions are educational recommendations and are not represented as actions implemented by NALCO. The interview questions are independently developed for educational learning and preparation and are not claimed to be actual NALCO recruitment interview questions. Real manufacturing decisions must follow applicable product specifications, approved procedures, safety requirements, and authorised engineering assessments.